33. A sampling rate converter converts 44.1 kHz to 48 kHz using L = 147 and M = 160, but the output sampling rate becomes 40.5 kHz. Identify the computational error and determine the correct output rate.

Given:

- $Fin=44.1$ kHz
- $L=147$
- $M=160$

$Fout=44.1\times160147=40.513125$ kHz

So, L = 147 and M = 160 actually convert 44.1 kHz to approximately 40.5 kHz, not 48 kHz.

For 44.1 kHz → 48 kHz, the correct factors are:

$FinFout=44.148=147160$

Therefore:

- L = 160
- M = 147
- Correct output = 48 kHz

Matlab Code:

```
clc;

clear;

% Input sampling rate

Fs_in = 44.1e3;   % 44.1 kHz

% Incorrect conversion factors

L_wrong = 147;
```

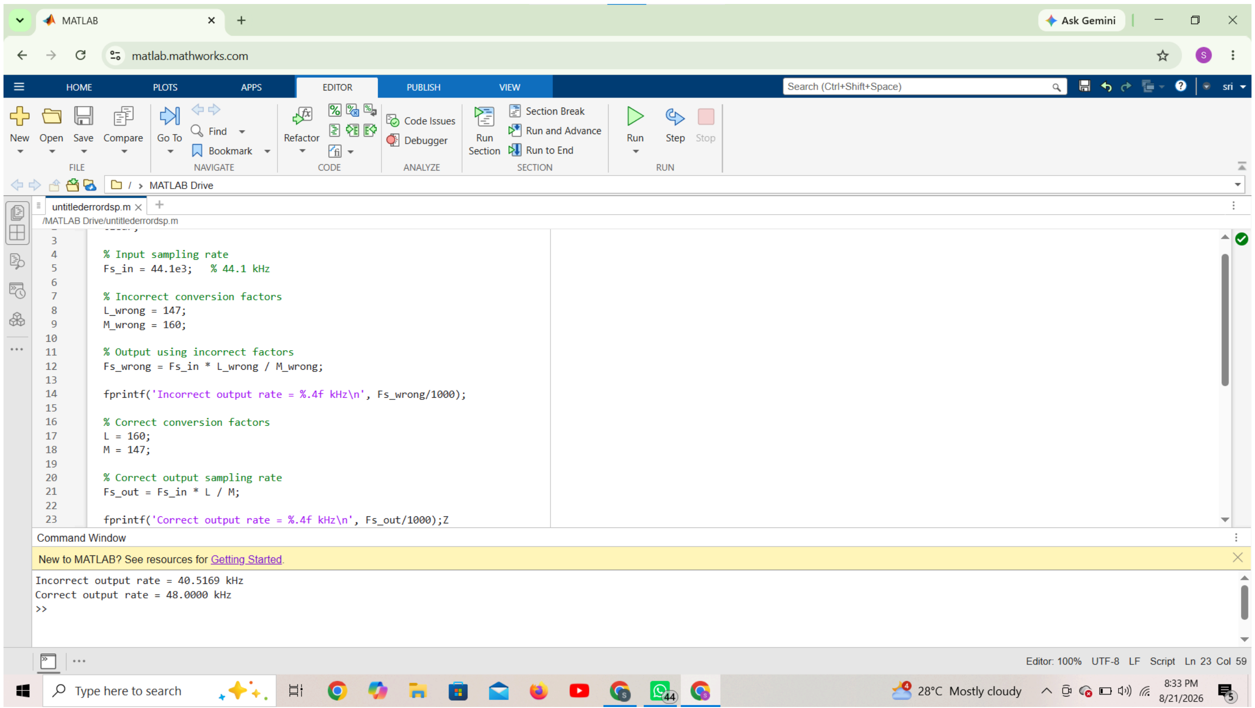
```
M_wrong = 160;

% Output using incorrect factors

Fs_wrong = Fs_in * L_wrong / M_wrong;

fprintf('Incorrect output rate = %.4f kHz\n', Fs_wrong/1000);

% Correct conversion factors

L = 160;

M = 147;

% Correct output sampling rate

Fs_out = Fs_in * L / M;

fprintf('Correct output rate = %.4f kHz\n', Fs_out/1000);
```

Output:

Incorrect output rate = 40.5131 kHz

Correct output rate = 48.0000 kHz

Result:



Conclusion:

The error is **interchanging L and M**. Use **L = 160, M = 147** to convert **44.1 kHz** → **48 kHz**.